Compiled on October 20, 2026.

Chi Zhao (Vector)

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Education

Saint Petersburg State University — Sept. 2021 – Jun. 2025
PhD in Applied Mathematics and Control Processes — *Saint Petersburg, Russia*
Thesis Title: "Modeling of binary opinion dynamics in social networks of complex configurations" — *Defended in Apr. 2025*

Saint Petersburg State University — Sept. 2019 – Jun. 2021
M.S. in Applied Mathematics and Informatics (GPA: 4.9/5.0) — *Saint Petersburg, Russia*

Beijing Institute of Technology — Jul. 3 – 28, 2017
ACM Summer Camp — *Beijing, China*

Yanan University — 2015 – 2019
B.S. in Computer Science (GPA: 3.1/4.0) — *Yanan, China*

Working Experience

Saint Petersburg State University — Dec. 2025 – Present
Assistant professor — *Saint Petersburg, Russia*

Huawei Inc. — Sep. 2021 – Dec. 2025
R&D Engineer — *Saint Petersburg, Russia*

Wisedu Inc. — Sep. 2016 – May. 2019
Software Engineer — *Yanan, China*

Project Experience

Russian Science Foundation Grant No. 26-71-100019 — 2026–2027
Senior Researcher — *Saint Petersburg, Russia*

- Hybrid computational methods and neural network surrogates for plasticity problems under high-velocity impact (Stage 1).

*** Professional model data governance/Valuable training data selection** — 2025
Project leader — *Saint Petersburg, Russia*

Project focused on improving model performance through data governance and optimized feature/training data selection.
Sampled 60M representative records from 1.8B wireless data using custom diversity sampling algorithms.
Boosted model accuracy by 22% using custom sampling and anomaly detection algorithms.
Accelerated training by 10% via custom feature importance algorithms that removed 20% redundant features.

*** Internal columnar storage algorithm** — 2023–2024
R&D Engineer — *Saint Petersburg, Russia*

High-performance, highly flexible columnar storage lossless compression algorithm
The algorithm will be commercialized to store base station data.
LTE (4G) data 30% CR, NR (5G) data 40% CR, without loss any performance.

Russian Science Foundation Grant No. 22-21-00346 — 2023
Researcher — *Saint Petersburg, Russia*

Completed simulation of two-layer opinion dynamics models, focusing on expressed and private opinions.

*** Elastic expansion algorithm for internal distributed file storage systems** — 2023
R&D Engineer — *Saint Petersburg, Russia*

Enabled seamless HOFS multi-node expansion using HRW Hashing with minimal data movement and no performance loss.
Improved system reliability by ensuring data access continuity even after hash ring loss.

*** Packet/Router data compression** — 2023
R&D Engineer — *Saint Petersburg, Russia*

This project achieved 85% CR without loss (Save 7x disk usage).

*** Wireless data compression** 2021–2022
R&D Engineer *Saint Petersburg,Russia*

This project achieved 96% CR through lossy compression.

*** SparkSQL query optimization** 2021
R&D Engineer *Saint Petersburg, Russia*

This project speeds up query execution by 50% through predicate pushdown and optimization of data structures.

Patent / Copyright Certificates

A program for morphological analysis and learning of the Russian language (RusLearn) 2025
Russian Computer Software Copyright Registration Certificate *NO. 2025696456*

A program for modeling the opinion dynamics in two-layer networks 2023
Russian Computer Software Copyright Registration Certificate *NO. 2023661532*

Online News Classification System Based on Convolutional Neural Network 2018
Chinese Computer Software Copyright Registration Certificate *NO. 2831192*

Open Source Projects

ShapG | *python · feature importance algorithm · centrality measures*
- A Python package for feature importance algorithms based on Shapley values.
- This package can also be used to calculate the centrality in the graph.

Helium | *Rust · columnar compression · DataFusion · SQL*
- A Rust columnar compression framework with composable, strongly typed encoding pipelines and JSON-driven schemas.
- Supports column pruning and SQL queries via Apache DataFusion, with stripe-level filter pushdown.

Skills

Programming: Python · C/C++ · Go · Rust · Matlab/Octave · Julia · R · SQL · LaTeX
ML Libraries: Tensorflow · PyTorch · Keras · Scikit-Learn
Developer Tools: Git · Docker · Google Cloud Platform
AI-assisted Development: Advanced use of Claude and OpenAI Codex for task decomposition, context management, and output verification.
OS: Windows · MacOS · Arch Linux
Languages: Chinese (native), and English (professional working proficiency)

Research interests

Graph Algorithms · Centrality Measures · Machine learning · Explainable Artificial Intelligence · Probability Theory · Statistics
Data Compression · Coding Theory · Time Series Analysis · Optimization · Stochastic Modeling · Stochastic Processes

Teaching

Optimization, Forecasting and Artificial Intelligence in Industry Saint-Petersburg, Russia
Lecturer *2026*

Deep learning Saint-Petersburg, Russia
Lecturer *2026*

Advance Economics Saint-Petersburg, Russia
Lecturer *2025*

Applied Statistics in R Saint-Petersburg, Russia
Teaching assistant *2024*

Statistical Decisions and Econometrics Saint-Petersburg, Russia
Teaching assistant *2022*

Community service

Reviewer for journals and conferences
- Engineering Applications of Artificial Intelligence (EAAI)
- Group Decision and Negotiation (GDN)
- Dynamic Games and Applications (DGAA)
- Humanities and Social Sciences Communications (HSSCOMMS)
- International Conference On Computational Optimization (ICOMP)

Selected Publications

1. Zhang Y., **Zhao C.**, Zhao S., Xu Z., Huang F., Petrov Yu. V. Physics-verified operator-learning-assisted inverse parameter identification of path-dependent plasticity // *International Journal of Plasticity.* – 2026. – Dec. – Vol. 207, 104835. **(Q1, IF (2025): 15.4)**

2. Liu J., **Zhao C.**, Parilina E. M. Feature importance methods for linear regression model. // *Applied Intelligence.* – 2026. – Jul. – Vol. 56, no. 10. – P. 332-356. **(Q2, IF (2025): 3.5)**

3. **Zhao C.**, Liu J., Parilina E. M. Strategic inputs: feature selection from game-theoretic perspective. // *arXiv preprint arXiv:2510.24982.* – 2025.

4. **Zhao C.**, Liu J., Parilina E. M. The Shapley Value Contribution to Explainable Artificial Intelligence: A Comprehensive Survey // *Dynamic Games and Applications.* – 2025. – Oct. – P. 1-38. **(Q2, IF (2025): 1.3)**

5. **Zhao C.**, Liu J., Parilina E. M. Complete-to-Sparse: A Novel Graph Construction Strategy for Efficient ShapG // *Mathematical Optimization Theory and Operations Research.* – Cham : Springer Nature Switzerland. – 2025. – P. 180–194.

6. **Zhao C.**, Liu J., Parilina E. M. ShapG: new feature importance method based on the Shapley value // *Engineering Applications of Artificial Intelligence.* – 2025. – May. – Vol. 148, 110409. **(Q1, IF: 8.0)**

7. **Zhao C.**, Parilina E. M. Centrality measures and opinion dynamics in two-layer networks with replica nodes // *Computers and Operations Research.* – 2026. – Jan. – Vol. 185, 107245. **(Q1, IF: 4.3)**

8. **Zhao C.**, Parilina E. M. Analysis of consensus time and winning rate in two-layer networks with hypocrisy of different structures // *Vestnik of Saint Petersburg University. Applied Mathematics. Computer Science. Control Processes.* – 2024. – Vol. 20, no. 2. – P. 170-192.

9. **Zhao C.**, Parilina E. M. Opinion Dynamics in Two-Layer Networks with Hypocrisy // *Journal of the Operations Research Society of China.* – 2024. – Mar. – Vol. 12, no. 1. – P. 109-132. **(Q2, IF (2025): 1.0)**

10. **Zhao C.**, Parilina E. M. Network Structure Properties and Opinion Dynamics in Two-Layer Networks with Hypocrisy // *Mathematical Optimization Theory and Operations Research.* – Cham : Springer Nature Switzerland. – 2024. – P. 300-314.

11. **Zhao C.**, Parilina E. M. Consensus time and winning rate based on simulations in two-layer networks with hypocrisy // *2023 7th Scientific School Dynamics of Complex Networks and their Applications (DCNA).* – 2023. – P. 68-71.

Conferences

21st International Symposium on Dynamic Games and Applications (ISDG 2026)	Istanbul, Turkey
Oral Presentation	*July. 20 - 23, 2026*
Mathematical Optimization Theory and Operations Research (MOTOR 2026)	Irkutsk, Russia
Oral Presentation	*July. 06 - 10, 2026*
International Conference On Computational Optimization (ICOMP 2025)	Abu Dhabi, UAE
Poster Presentation	*Oct. 17 - 19, 2025*
Mathematical Optimization Theory and Operations Research (MOTOR 2025)	Novosibirsk, Russia
Oral Presentation	*July. 07 - 11, 2025*
Game Theory and Management (GTM 2025)	Saint-Petersburg, Russia
Oral Presentation	*July. 2 - 4, 2025*
14th International Society of Dynamic Games (ISDG) Workshop	Yerevan, Armenia
Oral Presentation	*June. 11 - 13, 2025*
International Conference On Computational Optimization (ICOMP 2024)	Innopolis, Russia
Visitor	*Oct. 10 - 12, 2024*
Mathematical Optimization Theory and Operations Research (MOTOR 2024)	Omsk, Russia
Oral Presentation	*June. 30 - July. 06, 2024*
Game Theory and Management (GTM 2024)	Saint-Petersburg, Russia
Oral Presentation	*June. 26 - 28, 2024*
Dynamics of Complex Networks and their Applications (DCNA 2023)	Kaliningrad, Russia
Poster Presentation	*Sep. 18 - 20, 2023*
Game Theory and Management (GTM 2023)	Saint-Petersburg, Russia
Oral Presentation	*June. 28 - 30, 2023*

Honours & Awards

Name	*Placing*	*Scope*	*Awarder*	*Year*
3rd Prize in 2nd IMC Challenge	3rd	Global	Huawei Inc.	Aug. 2026
Outstanding contributor	-	Huawei Inc.	Huawei Chebyshev Research Center ADN AI Lab	Dec. 2025
AI4OPT Competition Best Paper	1st	ICOMP 2025	ICOMP 2025 Conference Organizing Committee	Oct. 2025
General Development Star	-	Huawei Inc.	General development department	Oct. 2024
General Development Star	-	Huawei Inc.	General development department	Jun. 2023
General Development Star	-	Huawei Inc.	General development department	Oct. 2022
President's Award (Team)	-	Huawei Inc.	MAE-M department	Dec. 2021
Diploma with distinction	-	University	Saint Petersburg State University	Jun. 2021
Excellent Graduation Thesis (Design)	-	University	Yanan University	Jun. 2019
Excellent Graduate	-	University	Yanan University	Jun. 2019
Merit Student Scholarship	-	University	Yanan University	Dec. 2018
China Software Cup Competition	3rd	China	China Software Cup Organizing Committee	Oct. 2018
Computer Design Competition	3rd	Northwest China	Northwest University (China)	May. 2018
Mathematical Contest in Modeling	2nd	Shaanxi	China Society for Industrial and Applied Mathematics	Dec. 2017
Mathematics Competition	3rd	Shaanxi	Chinese Mathematical Society	Nov. 2016